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How to check EEG data quality

This article covers the steps involved in assessing EEG data quality, and provides solutions for potential problems that may arise.

Good data not only accelerates your day-to-day research, it is also the deciding factor for the quality and reliability of your results. For an overview of the reasons for potential problems with EEG data, see the following article.

- The points listed below will help you get good EEG quality data. Make sure to read this through *before* collecting data!
- Once you have read the content of this article, we recommend you to do a short pilot test of all the points listed in this article.

You can check EEG quality data before, during, and after data collection.

BEFORE data collection: Once you have placed the EEG headset and correctly connected EEG electrodes and reference electrodes on a participant, go through the following points by looking at EEG channels on the software (more specifically on Enobio software called NIC if you use Enobio, or on iMotions ABM widget if you use ABM X10/24 headsets). It is also good practice to ensure that the headset is properly charged for use.

1. Impedance check (ABM, Enobio): Are impedance values green?

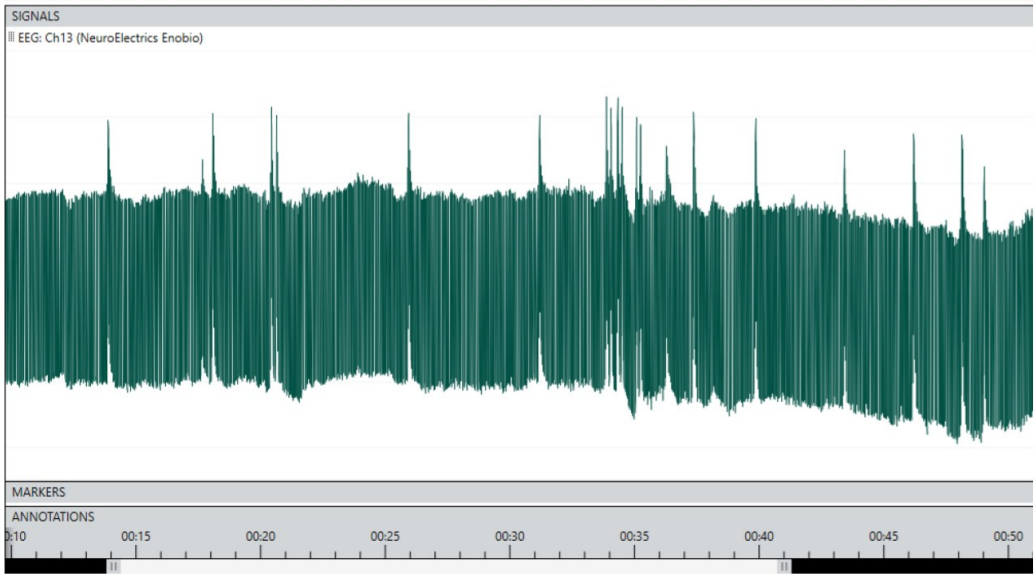
If only a number of the channels are shown in red, make sure there is a good contact between those EEG electrodes and the scalp of the participant, setting hair aside if you can. Gel can help you here to improve impedance values in an Enobio headset or an ActiCHamp headset, and conductive cream in an ABM headset. If this still does not work, try replacing the electrode/s. If the problem persists, please try to swap cables across two electrodes (one behaving well and one bad) to see if the faulty/noisy signal changes or remains at the same lead.

If all channels are at fault or have red impedances, make sure the skin where the reference electrode will be placed is clean and make sure to correctly place the reference electrodes on the recommended locations (mastoid processes and/or earlobe depending on the brand). If after completing this step, you still observe red impedances, read the following point.

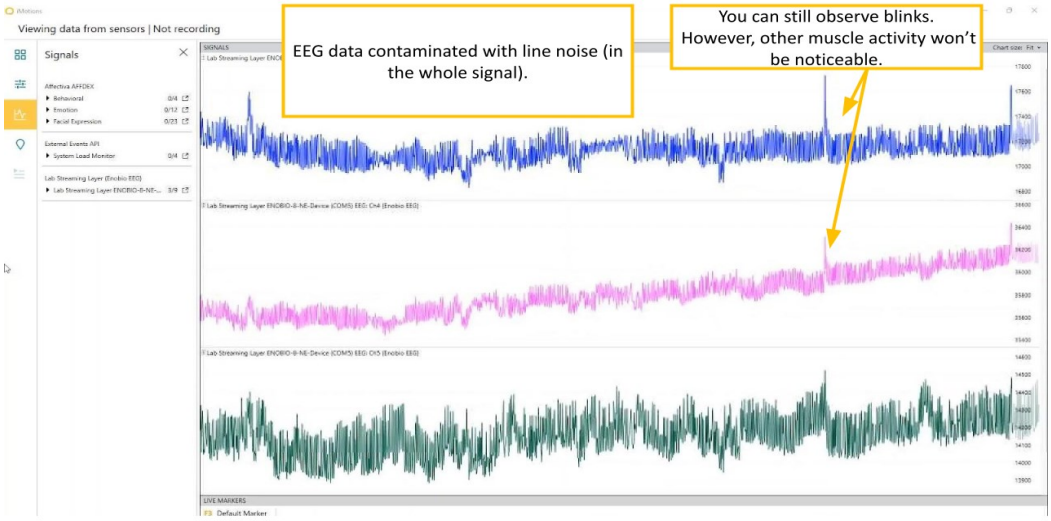
2. Are the PC and the monitor(s) connected to the wall grounded sockets through grounded plugs?

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Line noise is introduced in physiological signals by the electrical network. Using grounded plugs and sockets will help you reduce it. Line noise sometimes looks like this:

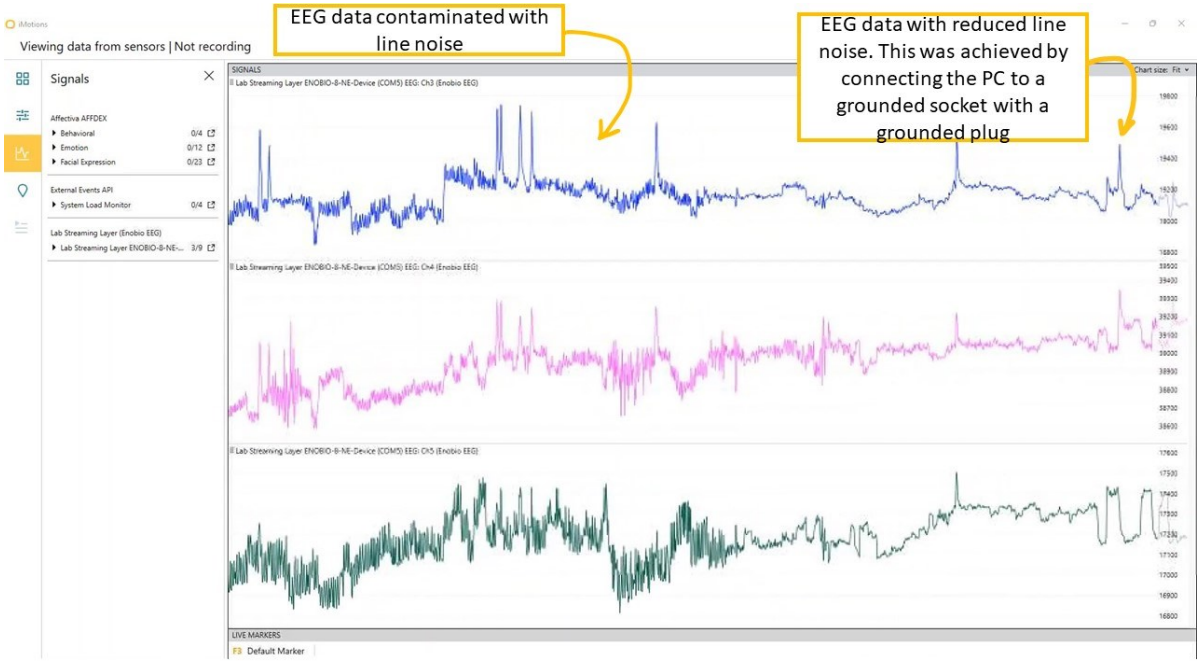


If the disruption is a bit less pronounced, it can look like this (spiky EEG data but you can identify some events apart from line noise):



After connecting the computer and monitor(s) as described in this point, the signal will look like the second half (right-hand side) of the picture below:

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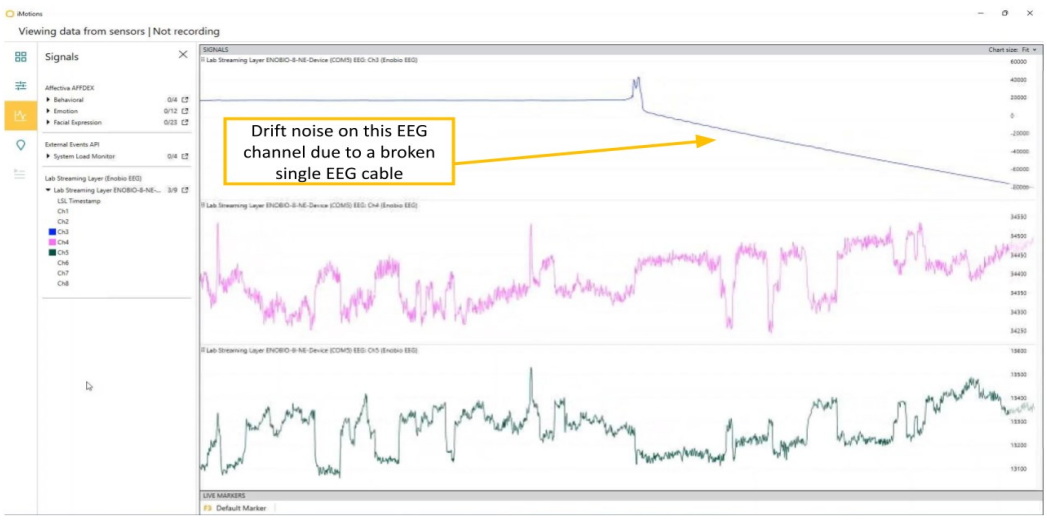


Line noise (50 Hz or 60 Hz depending on the country in which EEG data collection took place) is shown on the first half of the above recording.

3. Do you see drift noise on a specific channel?

Drift noise happens when there is either a poor connection or a worn-out EEG lead/cable. For the former, add some gel (for Enobio or Actichamp headsets) or conductive cream (for ABM headsets) to improve the contact with it (note that if you are using drytrodes with an Enobio headset then it is not applicable to add conductive gel). For the latter, try to swap EEG leads across two electrodes (one behaving well and one bad) to see if the faulty/noisy signal changes or remains at the same lead. If drift noise persists, try replacing the electrode/s and/or EEG cable lead (or simply set the EEG cable aside and stop using it for your data collection).

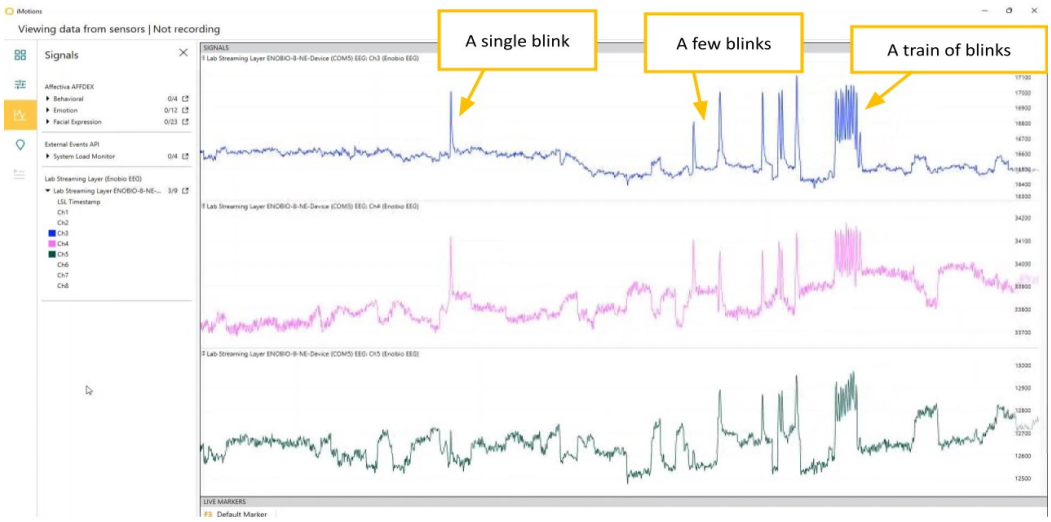
Drift noise can look like the following:

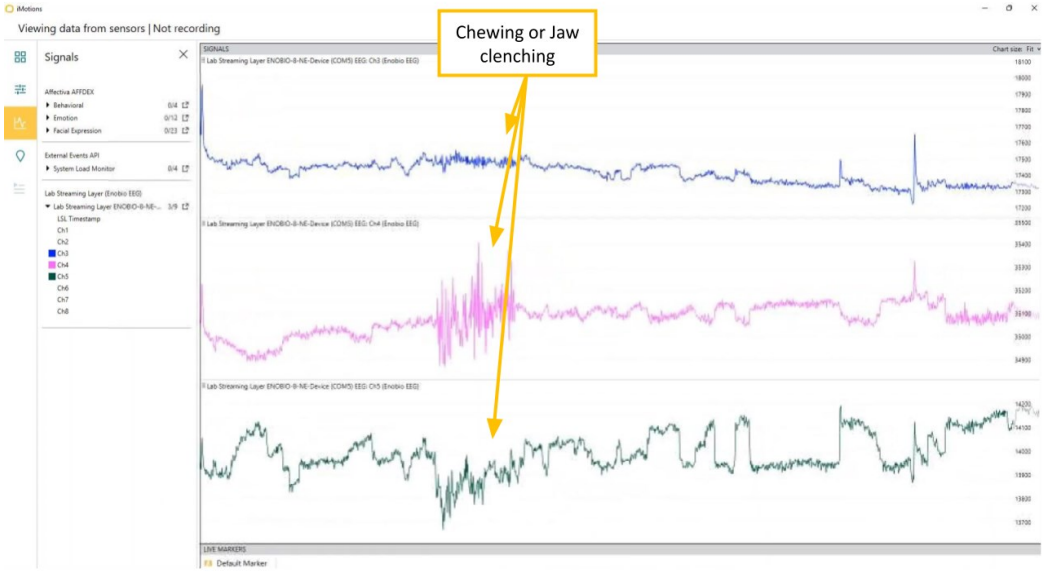


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4. Verification of blinks and other muscle artifacts on the sensor preview

Observing how EEG data looks like when each participant blinks a few times or presses their teeth (teeth clenching) or furrows their brows helps to confirm EEG data looks as expected, in other words, if the conditions for good EEG data are met, you will be able to observe and notice artifacts when they happen. It can be helpful to ask the participant to blink, press their teeth, or frown prior to data collection to see if this noise is expressed in the signal.





If you can see blinks in frontal electrodes (for example F3, F4, F7 or F8) and you can see muscle activity (in frontal electrodes if the participants try to furrow their brows, or in central electrodes like C3 and C4 if participants try to press their teeth), then this is a sign of having a good setup. If you don't observe these 2 artifacts, we suggest you revise the points above.

A sum-up of these 4 steps is shown below:

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DURING data collection:

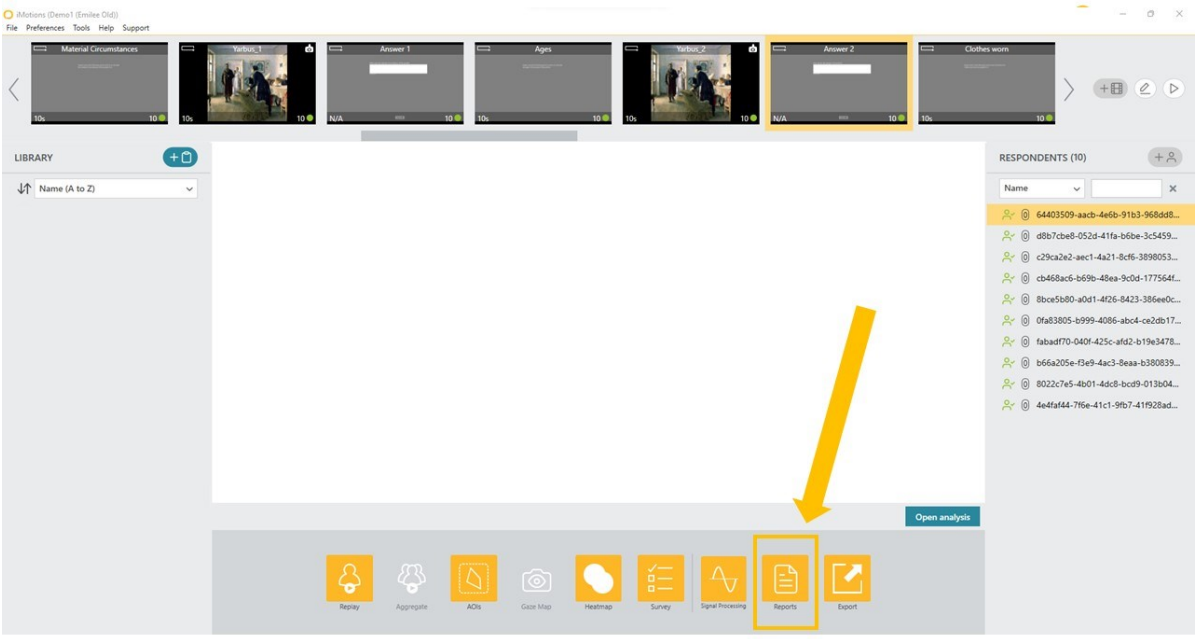
Make sure there is no surrounding noise interfering with the experiment and with the respondent's attention.

If the examiner is able to be close to the respondent PC (dependent on the experiment), the examiner can use live markers to track down the exact moments in which noisy EEG data is produced (e.g. when a respondent sneezes or coughs or when the examiner happens to observe drift noise). Using those live markers will allow you to keep track of the many events happening during data collection so that you can see later on how the artifact rejection tool excludes those intervals of time where noise occurred. In general, this is a good and time-saving methodology for future experiments. If the examiner does not have access to the respondent PC, you will need to skip this point. This process is recommended, but not essential to data collection, and you may decide to skip this step.

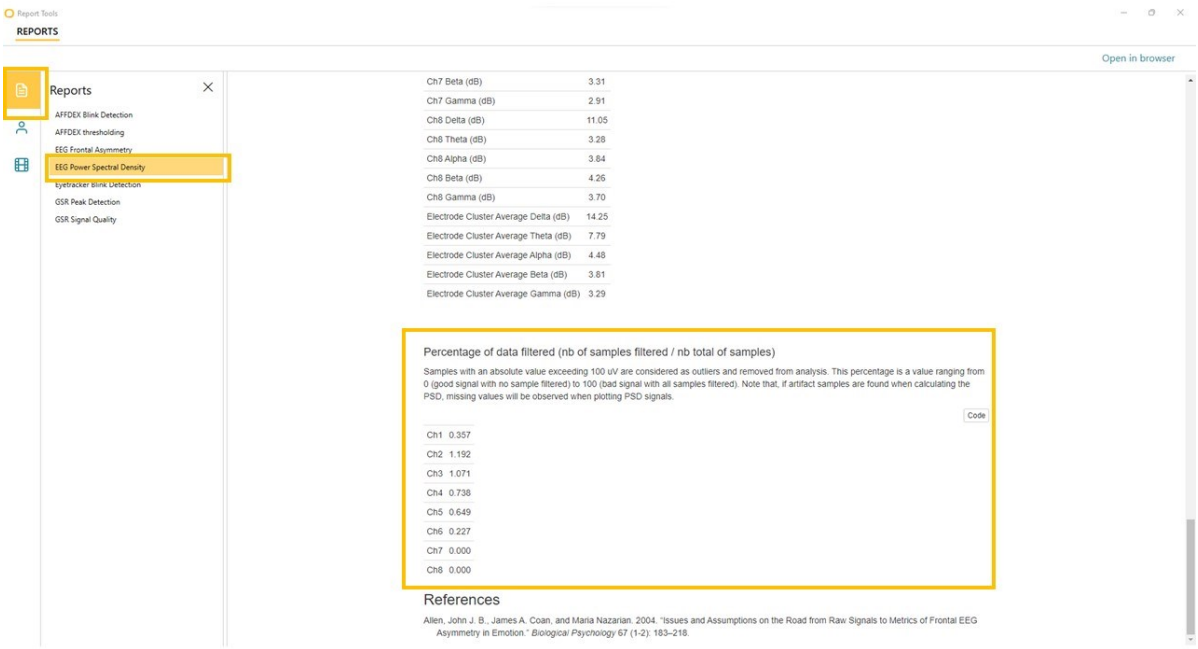
AFTER data collection:

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Once you have collected EEG data and finished one single experiment with a respondent, you can evaluate the EEG quality. To achieve that, you will create an Analysis in the study. After that, you can click on the Signal Processing button and run the EEG analysis that you would like (Frontal Asymmetry or Power Spectral Density). You can apply artifact rejection threshold (explained in the Parameters section in the Frontal Alpha Asymmetry article or Power Spectral Density article) which will automatically remove data points that have an amplitude equal to or higher than the chosen threshold (by default 120 uV but you can modify it according to your criteria). Click the "Reports" button in iMotions to see (scrolling down the Reports) the percentage of data that has been removed with this tool. We encourage you to do this every time you collect data on each respondent, so that you have an understanding of what the EEG quality is before completing your whole project.



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